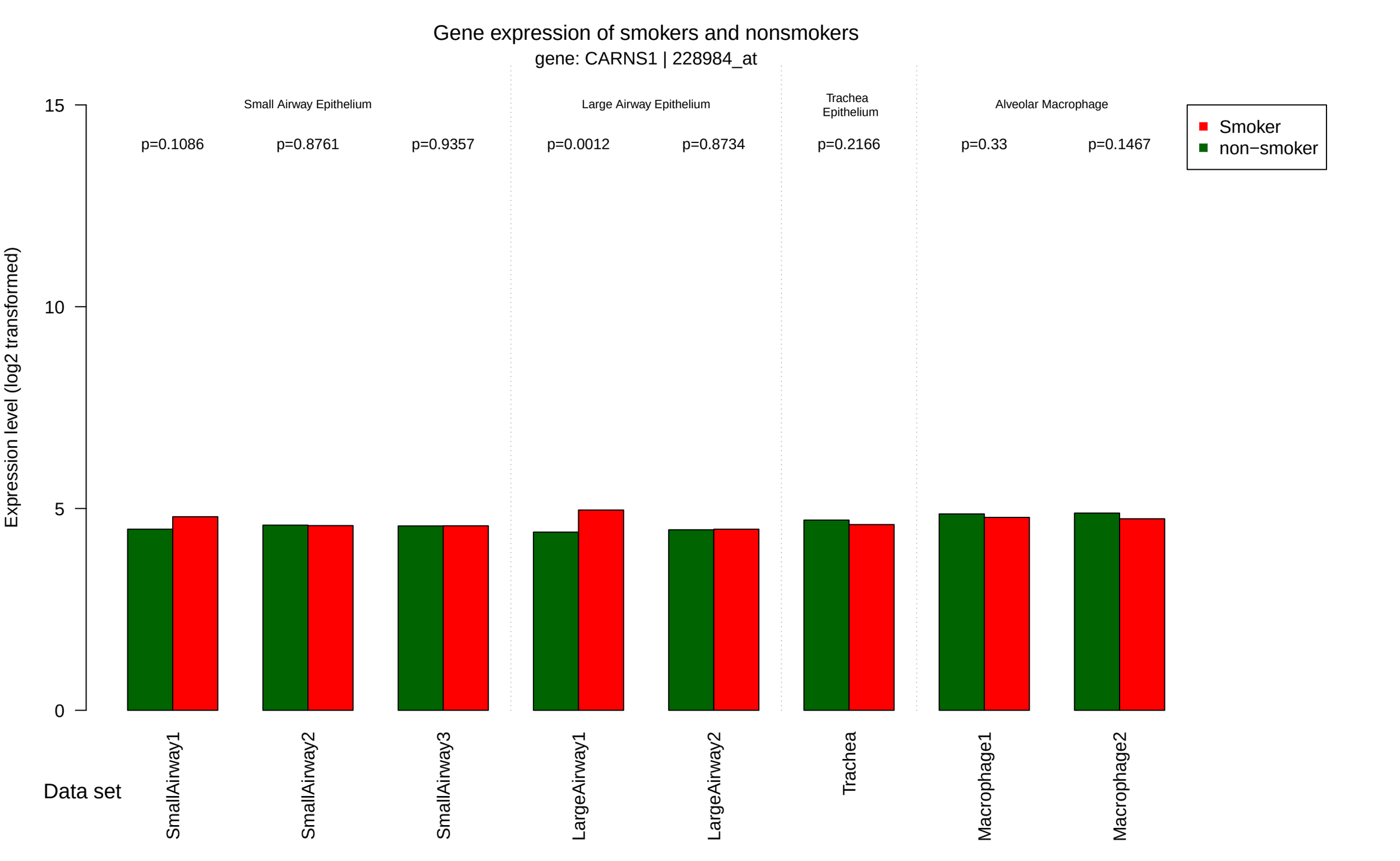
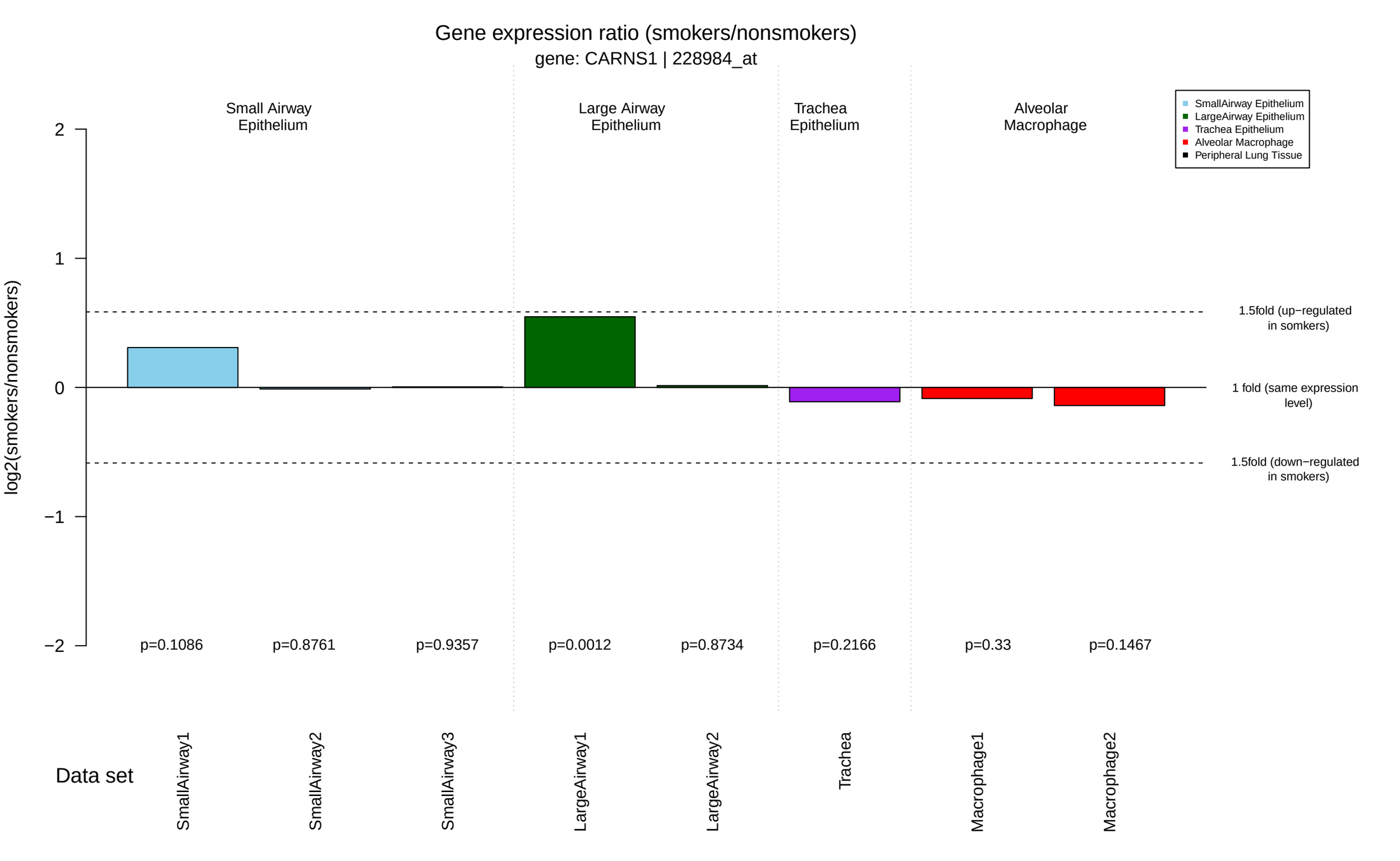
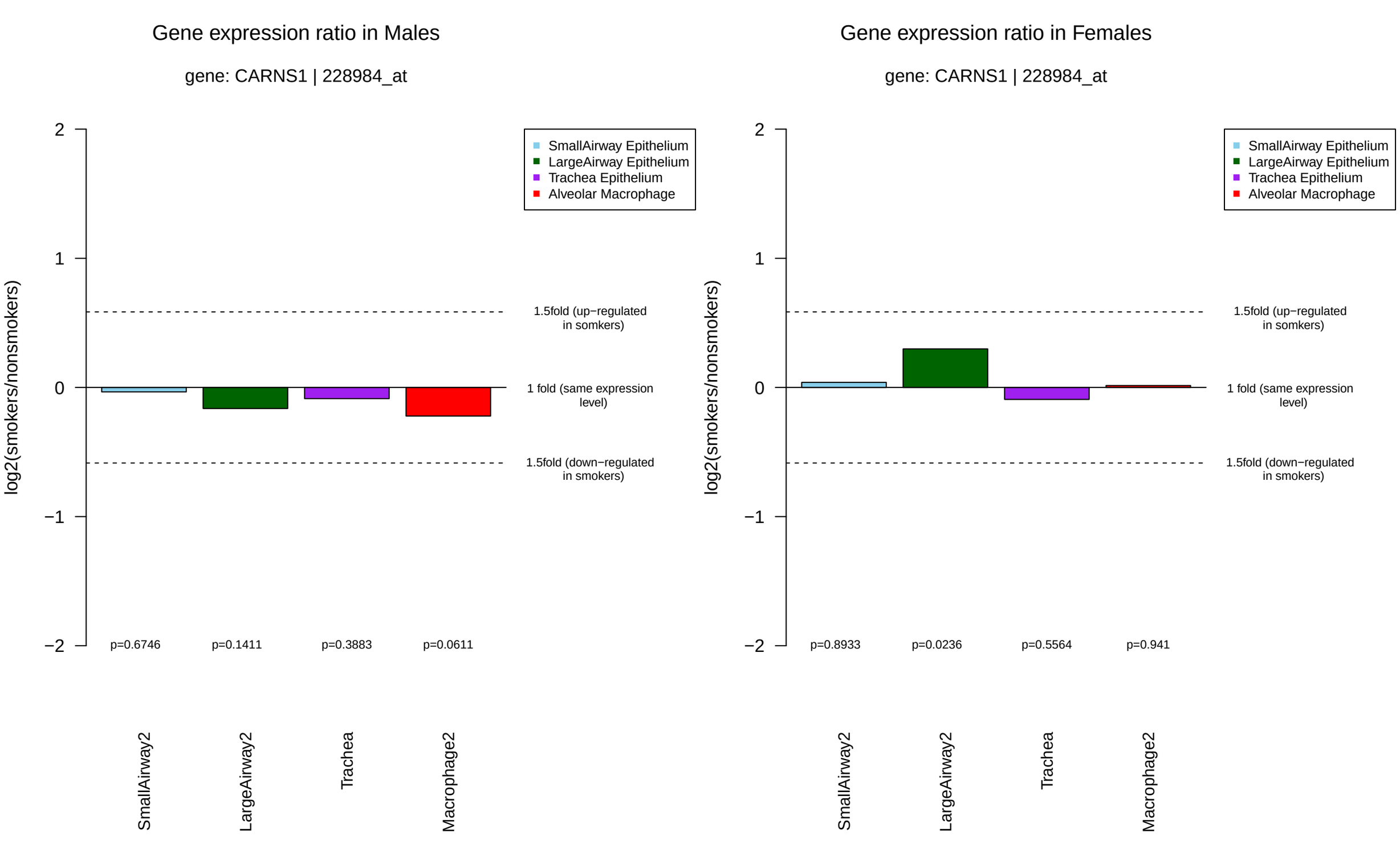
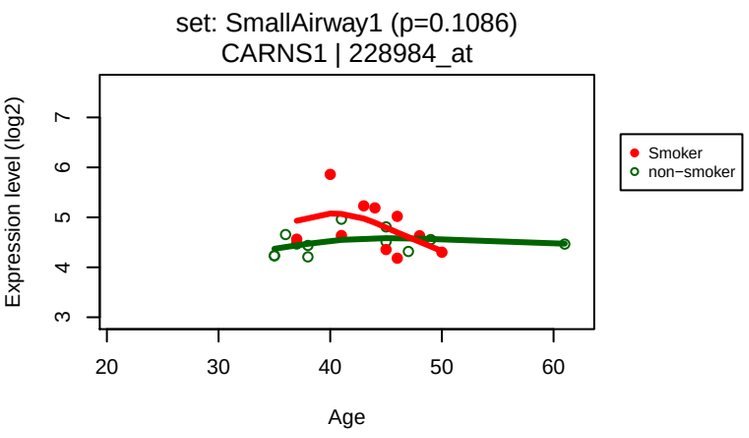
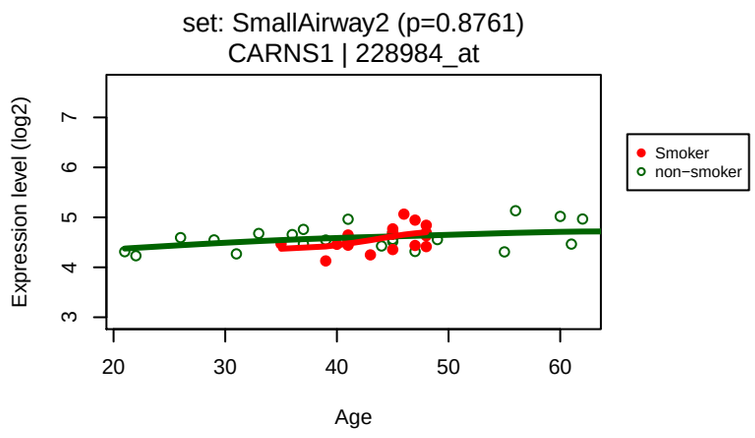
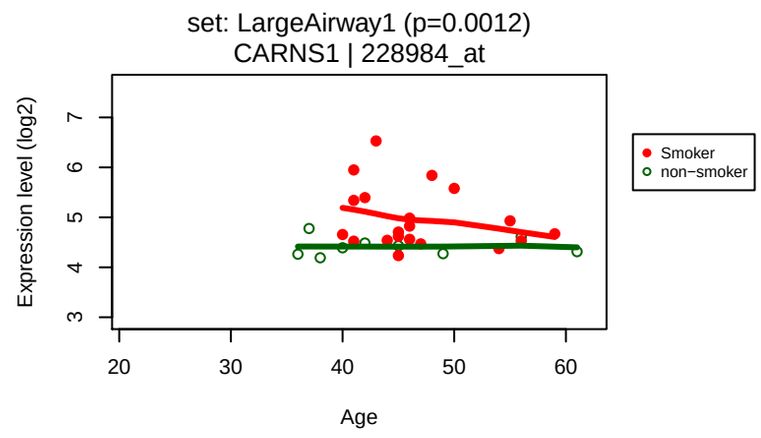

Gene expression of smokers and nonsmokers
gene: CARNS1 | 228984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1086
p=0.8761
p=0.9357
p=0.0012
p=0.8734
p=0.2166
p=0.33
p=0.1467
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

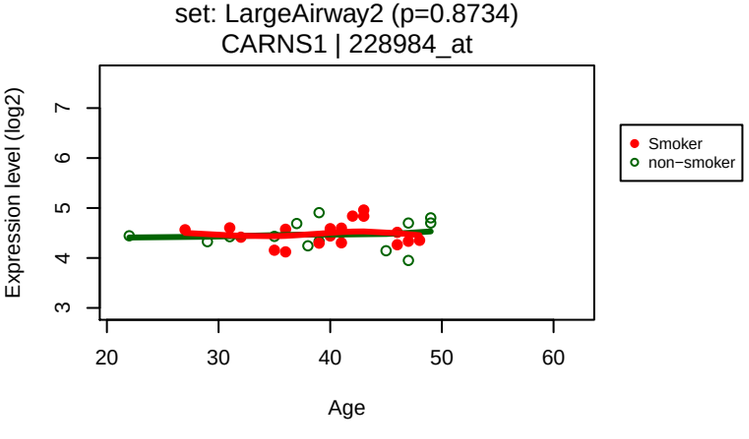

Gene expression ratio (smokers/nonsmokers)
gene: CARNS1 | 228984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1086
p=0.8761
p=0.9357
p=0.0012
p=0.8734
p=0.2166
p=0.33
p=0.1467
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

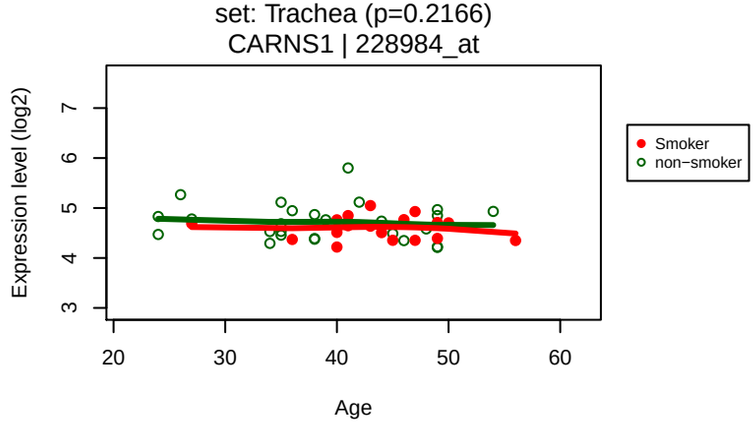

Gene expression ratio in Males
gene: CARNS1 | 228984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6746
p=0.1411
p=0.3883
p=0.0611
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CARNS1 | 228984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8933
p=0.0236
p=0.5564
p=0.941
SmallAirway2
LargeAirway2
Trachea
Macrophage2

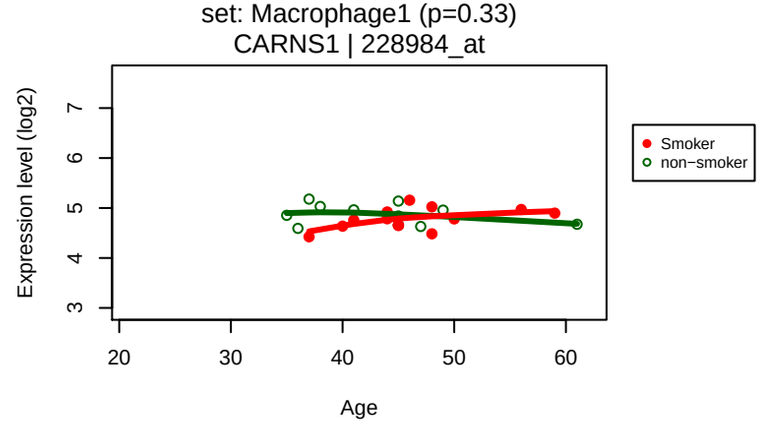

set: SmallAirway1 (p=0.1086)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker

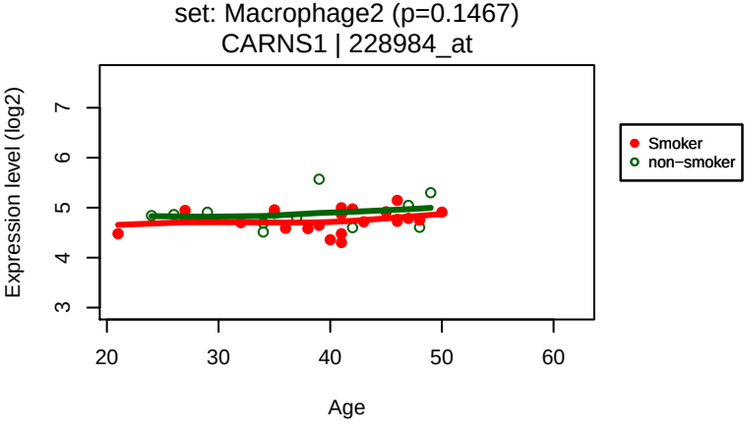

set: SmallAirway2 (p=0.8761)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker

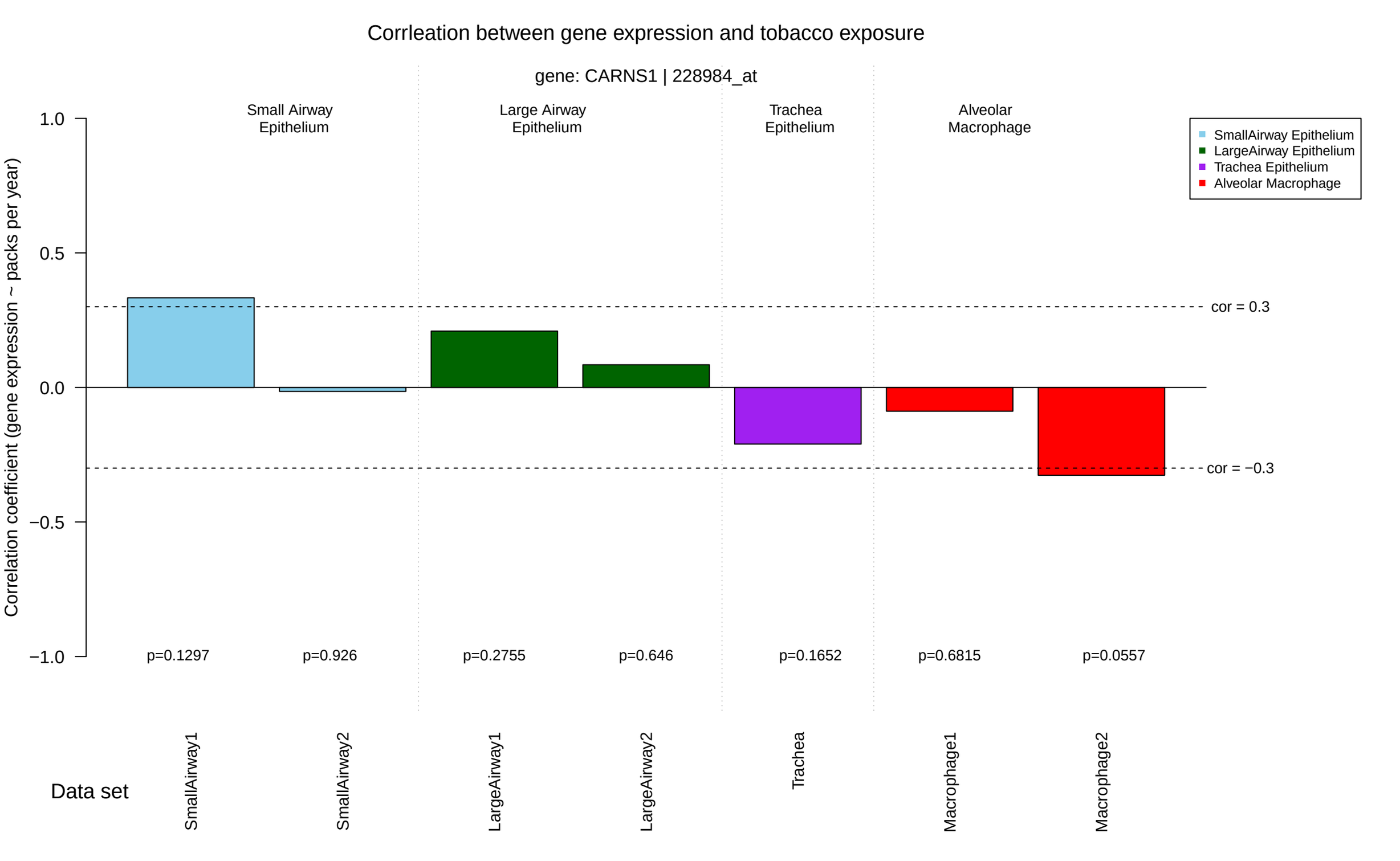

set: LargeAirway1 (p=0.0012)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8734)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2166)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.33)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1467)
CARNS1 | 228984_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CARNS1 | 228984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1297
p=0.926
p=0.2755
p=0.646
p=0.1652
p=0.6815
p=0.0557
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2